Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details:

Frontend: Soumitri Roy, Tanu Kashyap

Backend: Shabira Begam

Database: Sneha Lahudkar, Thrisha D

AI integration: Soumitri Roy, Shabira Begam

Batch Number 01: Enterprise learning platform

Start date : 29-6-2026

Name: Tanu Kashyap

Internship Duration: 40-day (12 Weeks)

1. Project Title

Enterprise learning platform

2. Project Objective

Enterprise learning platform is an AI-powered full-stack learning and workforce platform designed to bridge the gap between education and industry. The main objectives of this project are:

- **To provide users with an interactive AI study buddy** for personalized learning support and automated concept explanations.
- **To analyze student profile data** and provide insights into learning progress, skill acquisition, and career readiness.
- **To match students with career opportunities** using an intelligent job recommendation engine based on skill analysis.
- **To facilitate candidate recruitment workflows** with applicant tracking, candidate evaluation, and team management dashboards for recruiters.
- **To create administrative panels** that monitor user management, platform diagnostics, complaint tracking, and global analytics in an easy-to-understand format.

3. Project description in detail

Enterprise learning platform is a web-based AI-powered learning and workforce platform developed using a modern decoupled architecture with a React 19 frontend and a Java Spring Boot backend. The platform allows users to manage their education, build industry skills, and navigate recruitment pipelines. The system consists of multiple portals and modules that work together to provide a complete learning and talent acquisition experience.

Landing Page:

The landing page introduces users to the platform and highlights the key features of SkillSphere. It provides details on learning pathways, recruiter workspaces, and admin interfaces, encouraging users to register or log in.

Authentication System:

The system includes a secure login and registration mechanism using role-based JWT (JSON Web Token) authentication and Google OAuth 2.0, allowing students, recruiters, and administrators to safely access their respective dashboards.

Student Dashboard:

The dashboard is the main interface where students can view their learning insights. It displays interactive statistics about active courses, daily quest completions, flashcard reviews, and skill analytics.

AI Study Buddy:

The chatbot interacts with students to provide academic assistance, explain complex engineering concepts, and generate personalized learning paths. It acts as a virtual assistant that guides users throughout their educational journey.

Intelligent Recommendation Engine:

The platform uses recommendation models to match student profiles with active recruiter job listings. The system evaluates a set of features including:

- Student skill tags
- Completed course certificates
- Resume keywords
- Job opportunity descriptions
- Recruiter compensation and location properties

The recommendations help students discover the best career matches for their skill sets.

Workforce Portal:

The recruiter portal allows organizations to post job listings, evaluate candidate qualifications, review resumes, and track applicants through a structured hiring workflow (Applied, Screening, Interviewing, Offered, Rejected).

Admin Panel:

An admin panel allows administrators to monitor system diagnostics (latency and load), manage user accounts, update course syllabi, resolve complaint tickets, and review platform engagement reports.

3. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project objective is decided, making github repository	GitHub repository created, finalising teams with the project overview
Week 2	Tech stack evaluation, architecture planning	Spring Boot and React 19 (Vite) setup completed, learning JPA/Hibernate configurations
Week 3	Setup user authentication, JWT tokens, and OAuth	Done with verified signin / login with JWT sessions and Google OAuth 2.0
Week 4	Database design, JPA entity mappings, and SQL commands	Schema designed mapping User, Course, and Opportunity tables in MySQL
Week 5	API endpoints setup, connecting frontend to backend services	REST API endpoints developed in Spring Boot and configured Axios interceptors in React
Week 6	Student Dashboard and Student Profile interface design	Implemented student dashboard layout, stats cards, and skill profile editing inputs
Week 7	Learning Hub, Course management, and video modules integration	Set up course search catalogs and integrated YouTube Data API for video lectures
Week 8	Workforce Portal and Recruiter Dashboard creation	Completed job posting wizard form and applicant listing views for recruiters
Week 9	Hiring Pipeline State Machine implementation	Configured candidate pipeline state transitions (Applied, Screening, Interviewing, Offered, Rejected)
Week 10	Google Gemini AI API integration	Deployed AI Study Buddy assistant chatbot and AI career roadmap generation
Week 11	Monaco Editor sandbox and Resume Builder integrations	Integrated Monaco Editor inside the Coding Arena page and built the resume compiler
Week 12	UI polishing, styles verification, final testing	theme polished across dashboards, JUnit test suites verified, project ready for evaluation

5a. Key Milestones

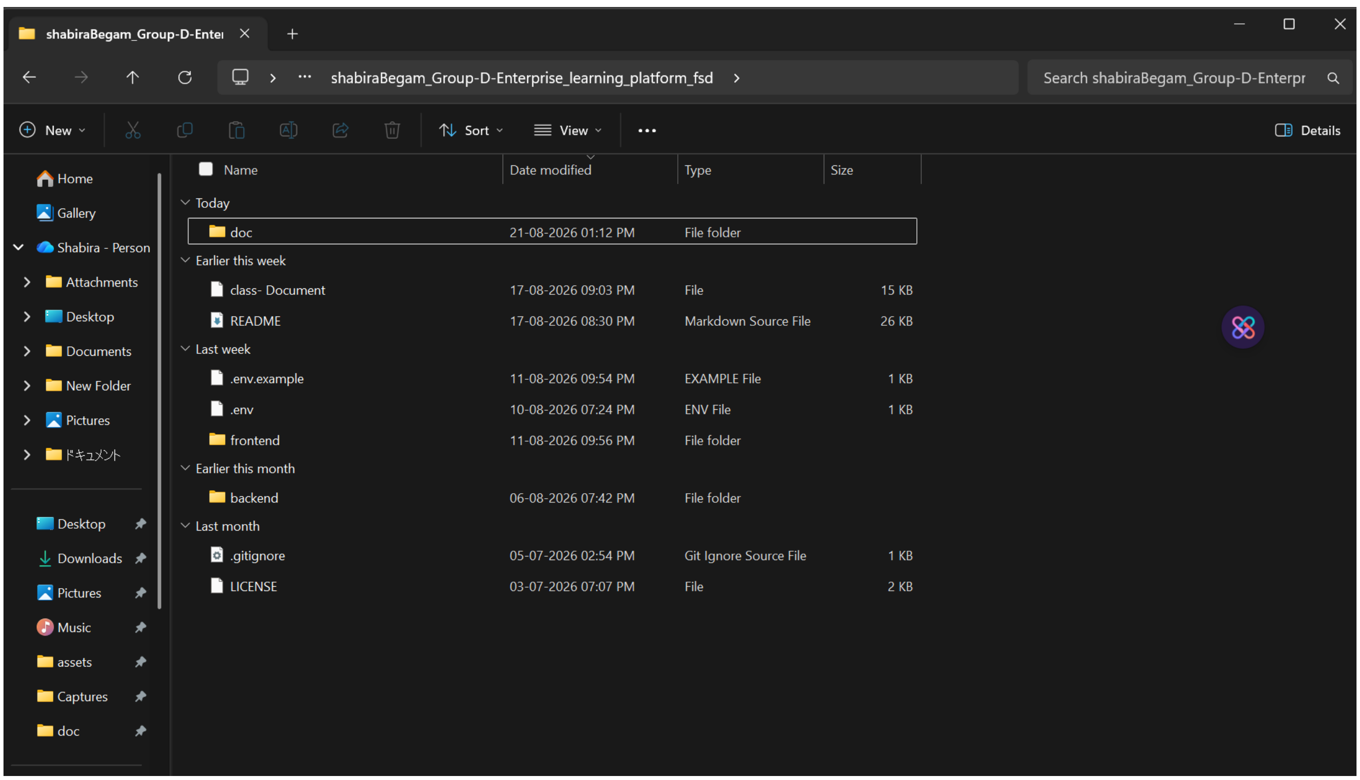
Milestone	Description	Date Achieved
Project Kickoff	Defined the project objectives, scope, system architecture, database schema, and roles for developing the Enterprise learning platform.	**29 June 2026**
Prototype / First Draft	Developed the initial working prototype, including the Spring Boot backend REST endpoints, React routing framework, and basic user authentication (JWT/Google OAuth).	**08 July 2026**
Mid-Term Review	Evaluated project progress, deployed Student and Recruiter dashboard interfaces, integrated course listing catalogs, and configured candidate application pipelines.	**23 july 2026**
Final Submission	Completed the fully functional Enterprise learning system, including the Gemini-powered AI Study Buddy, Monaco-based Coding Arena, and resume compilation wizards.	**13 Aug 2026**
Final Presentation	Demonstrated the working system highlighting features such as personalized AI career guidance, sandboxed code execution, recruiter pipeline tracking, and global admin analytics..	**1 Sep 2026**

5b. Project execution details

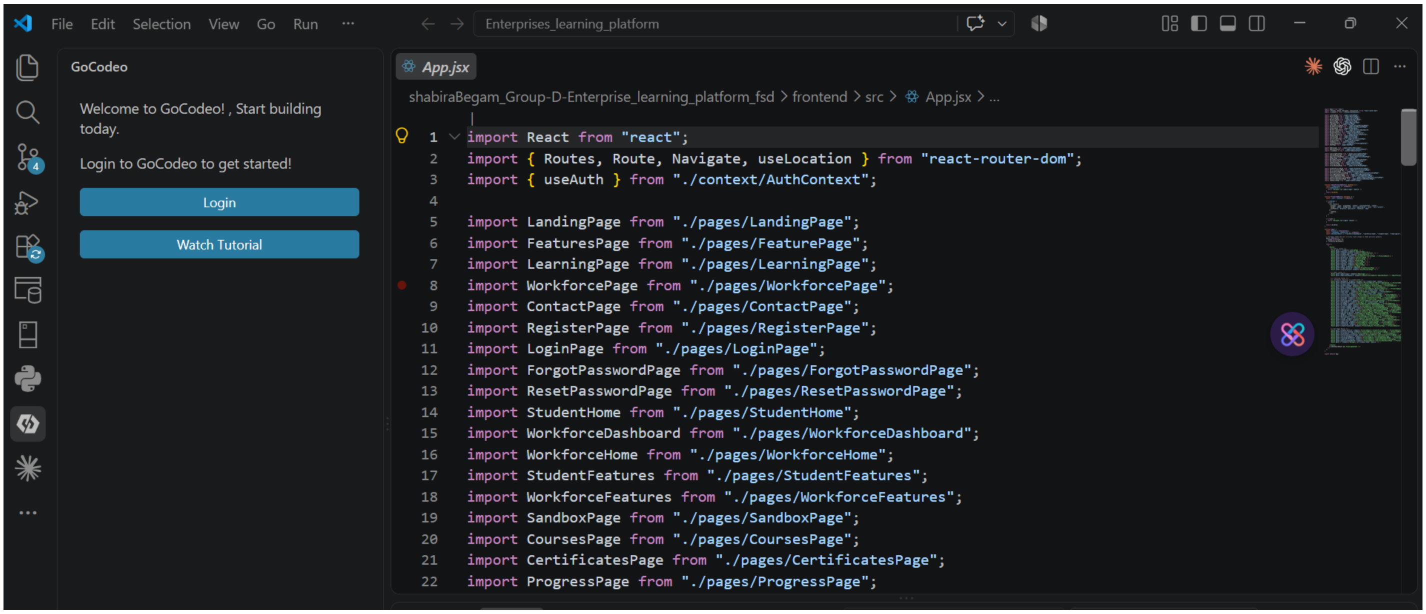
1. **Requirement Analysis:** The project started with identifying the main objectives of the Enterprise Learning Platform system, which included building an AI-powered learning and workforce platform that tracks student progress, facilitates recruiter pipelines, provides chatbot support, and generates skill-based job matches.
2. **System Architecture Design:** The architecture of the system was planned using a decoupled client-server model where the React 19 frontend interacts with the Spring Boot backend REST APIs. The backend processes user requests, communicates with the MySQL database, and integrates external API layers.
3. **Frontend Development:** The user interface was developed using React 19 and Vite. Multiple views were created, including the landing page, login/signup portals, student learning hub, recruiter dashboard, and admin diagnostics dashboard, utilizing a premium Glassmorphism design system.
4. **Backend Development:** The backend was implemented using Java 17 and Spring Boot (3.2.5). It handles role-based security configurations, manages JWT generation/validation, handles business transactions, and manages communication with external AI providers.
5. **Database Integration:** A MySQL 8.0 relational database was used to store user profiles, course materials, job opportunities, and application pipelines. Database operations were handled efficiently using Hibernate ORM and Spring Data JPA.

6. **AI Integration:** Google Gemini AI (gemini-1.5-flash) was integrated to power the conversational AI Study Buddy, auto-generate study guides, and review resumes. The YouTube Data API was integrated to dynamically fetch learning videos.
7. **Job Recommendation Engine Development:** A vector matching recommendation service was implemented on the backend. The service compares student skill sets against active job requirements to generate tailored career suggestions.
8. **Coding Arena Sandbox Development:** A browser-based code editing sandbox was built using Monaco Editor. The backend compiles and evaluates user code snippets against predefined assertions, returning outputs in real-time.
9. **Dashboard Analytics & Tools**: Interactive dashboards were designed to visualize student progress charts and recruiter applicant metrics. A custom resume builder was implemented, allowing users to compile and download resumes.
10. **Admin Panel and Testing**: An administrative panel was implemented to manage users, update courses, and trace server loads. The entire system was validated using JUnit and Mockito unit test suites to ensure smooth system integration.

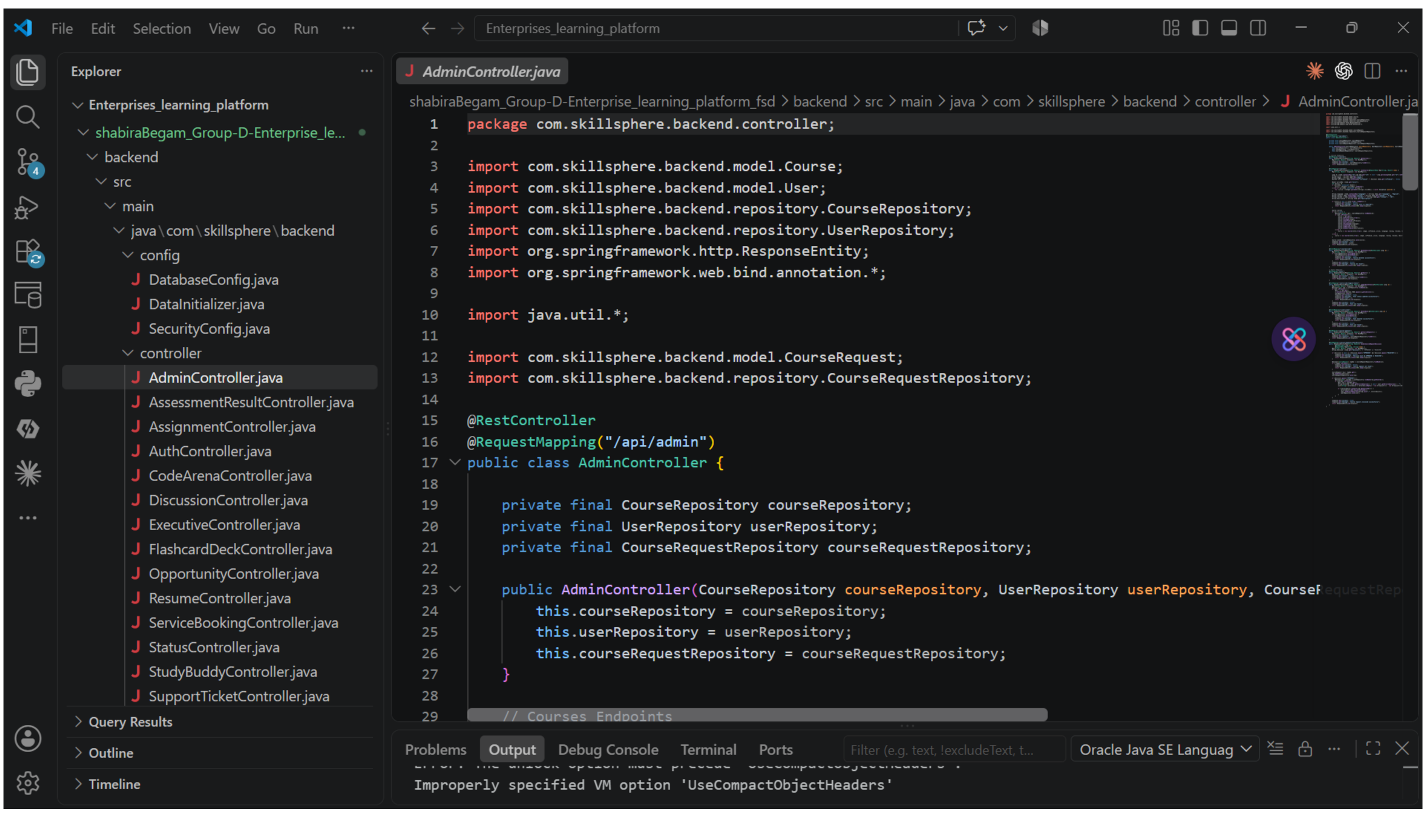
6. Snapshots / Screenshots



Opening the project folder containing all the main files Backend, Frontend and .env file.



Opening the Frontend file in VS Code to view in app.jsx main application code



Opening the Backend file in VS Code to view in java application code

```
Problems    Output    Debug Console    Terminal    Ports                                node - frontend

PS C:\Users\SHABIRA\Downloads\Enterprises_learning_platform> cd shabiraBegam_Group-D-Enterprise_learning_platform_fsd
>>
PS C:\Users\SHABIRA\Downloads\Enterprises_learning_platform\shabiraBegam_Group-D-Enterprise_learning_platform_fsd> cd fro
ntend
PS C:\Users\SHABIRA\Downloads\Enterprises_learning_platform\shabiraBegam_Group-D-Enterprise_learning_platform_fsd\fronten
d> npm run dev

> frontend@0.0.0 dev
> vite

1:21:09 pm [vite] (client) Re-optimizing dependencies because vite config has changed

  VITE v8.1.3  ready in 2937 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
1:21:11 pm [vite] (client) [optimizer] bundling dependencies...
```

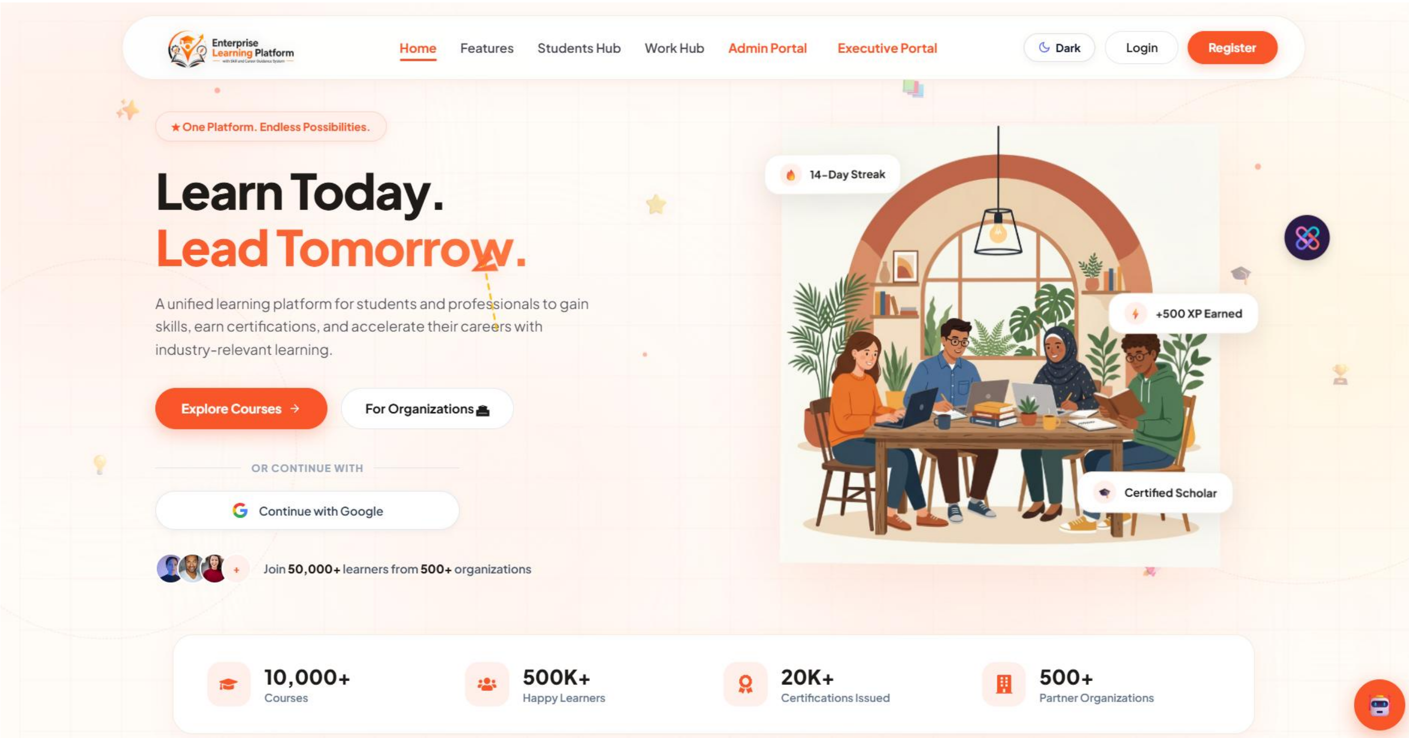
Running the frontend in the terminal of VS code.

```
PS C:\Users\SHABIRA\Downloads\Enterprises_learning_platform> cd shabiraBegam_Group-D-Enterprise_learning_
platform_fsd
>>
PS C:\Users\SHABIRA\Downloads\Enterprises_learning_platform\shabiraBegam_Group-D-Enterprise_learning_plat
form_fsd> cd backend
PS C:\Users\SHABIRA\Downloads\Enterprises_learning_platform\shabiraBegam_Group-D-Enterprise_learning_plat
form_fsd\backend> mvn spring-boot:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDe
finer (file:/C:/Users/SHABIRA/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.Hidden
ClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.skillsphere:backend >-----------------------
[INFO] Building skillsphere-backend-java 1.0.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:3.2.5:run (default-cli) > test-compile @ backend >>>
[INFO]
```

Running the Backend in the terminal of VS code.

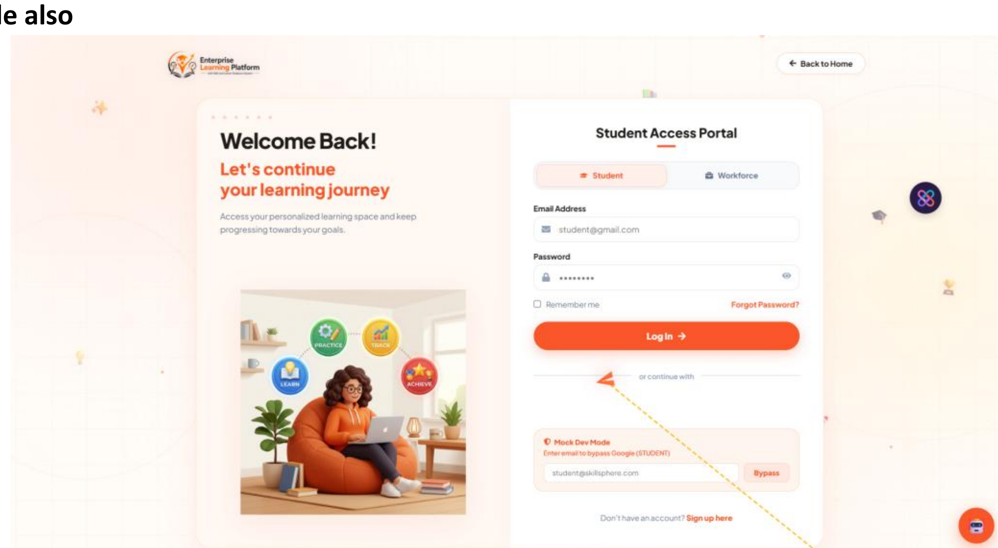
1. **Copying or clicking the local server link provided by react (** http://localhost:5173/**).**

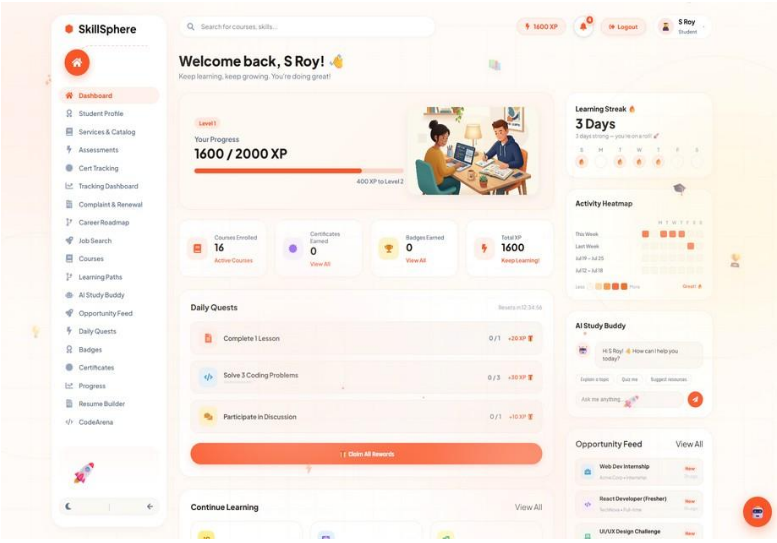
The Landing page of the Enterprises learning platform appears on the browser.

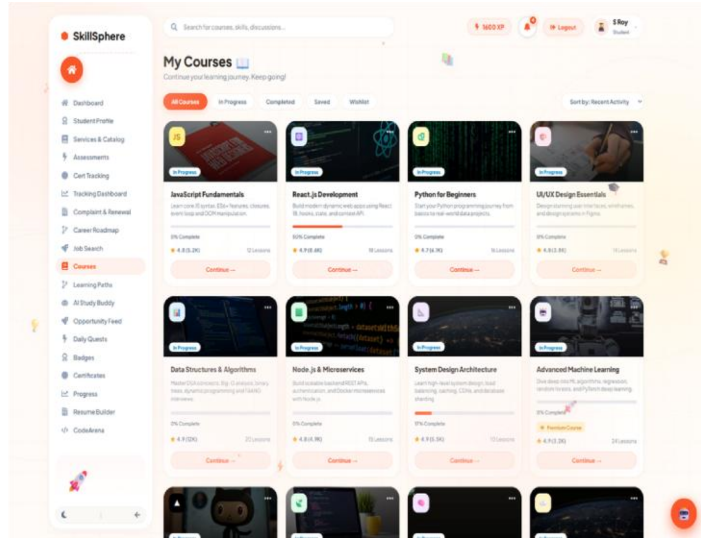


LANDING PAGE

3. Next page includes log in and also create a account for the new user or you can start your account with google also

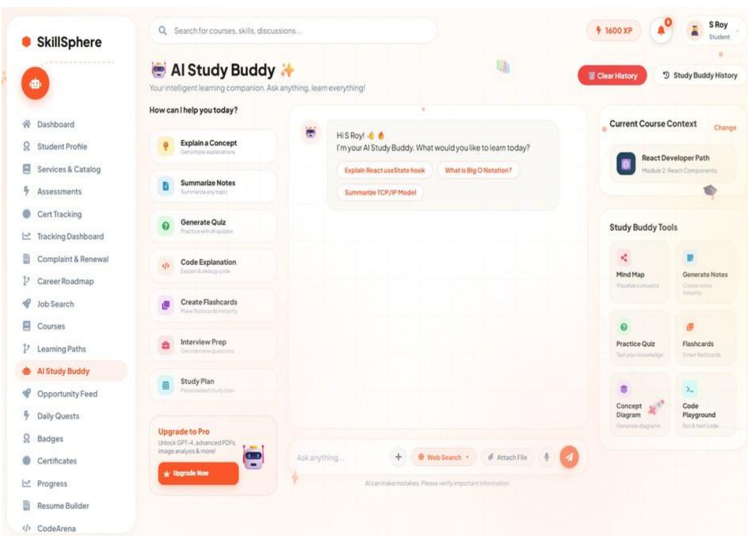


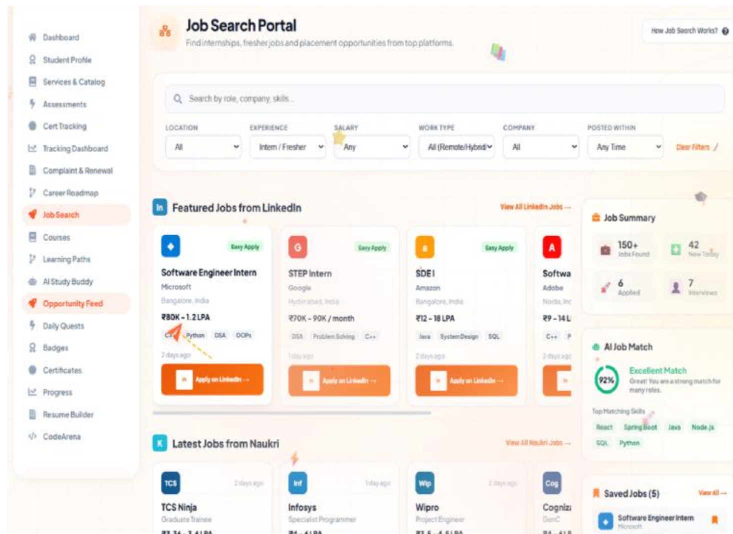




Student Dashboard

Courses Dashboard





Chatting with different coaches is also available in the application

7. Challenges Faced

During development of the Enterprise Learning Platform ,several technical and operational challenges were encountered:

1. **Integrating Generative AI with the Web Application:** One of the main challenges was connecting the Google Gemini API with the Spring Boot backend so that study resources, career roadmaps, and resume feedback could be generated dynamically based on student inputs.
2. **Designing a User-Friendly Dashboard:** Creating interactive and visually clear dashboards (for Students, Recruiters, and Admins) that display learning metrics, skill analytics, and recruitment pipelines in an understandable format required careful UI design and data visualization planning.

3. **Handling Failover and AI Response Generation:** Designing a multi-tiered fallback architecture to ensure the AI Study Buddy continues to function smoothly by switching to alternative free LLM gateways (such as Pollinations AI) or offline custom knowledge engines when API quotas or keys are unavailable.
4. **Database Management:** Managing user accounts, course data, coding challenges, and applications securely, and maintaining proper database connectivity with MySQL using Hibernate and Spring Data JPA, required careful implementation.
5. **Chatbot Integration:** Integrating the AI Study Buddy chatbot system and ensuring it responds correctly to user queries with appropriate formatting (like markdown headers and code blocks) was another challenge.
6. **Frontend and Backend Communication:** Ensuring smooth communication between the React frontend interface and Spring Boot backend through RESTful APIs and secure JWT headers required debugging and testing.
7. **System Testing and Debugging:** During development, several issues such as CORS policy blocks, database table conflicts, and authorization exceptions (JWT token expirations) were encountered and needed to be resolved through testing.
8. **Ensuring Smooth User Experience:** Making sure that all modules such as authentication, dashboards, AI chatbot, Coding Arena, and recruiter tracking portals worked seamlessly together was a significant challenge during the final integration phase.

8. Learnings & Skills Acquired

1. **Java Programming:** Understanding Java 17 was essential for developing the robust, object-oriented backend logic of the application and implementing core service architectures.
2. **Spring Boot Framework:** Knowledge of Spring Boot helped in building the web application backend, managing RESTful endpoints, handling request routing, and integrating different modules of the system.
3. **Frontend Development:** Skills in React 19, Vite, and modern CSS (with Glassmorphism and micro-animations) were required to design responsive, component-based, and user-friendly interfaces such as the dashboard portals, Coding Arena, and chatbot screens**.**
4. **Database Management & ORM:** Knowledge of MySQL, Hibernate, and Spring Data JPA was required to design relational database schemas, manage queries, and map complex relationships for users, courses, and job applications efficiently.
5. **Generative AI Integration:** Understanding how to integrate Google Gemini AI models (gemini-1.5-flash/2.0-flash) was necessary to build the conversational AI Study Buddy, dynamic career roadmaps, and automated resume reviews.
6. **Web Security & Authentication:** Skills in implementing secure stateless authentication using JSON Web Tokens (JWT), role-based access control (RBAC) in Spring Security, and Google OAuth 2.0 were acquired to safeguard user data and endpoints.
7. **API Integration & Failover Strategies:** Knowledge of external APIs, such as the YouTube Data API for pulling learning resources and Pollinations AI for failover mechanisms, helped build resilient, data-rich features.
8. **Problem Solving and Debugging:** Strong debugging and problem-solving skills were needed to resolve cross-origin resource sharing (CORS) policy blocks, manage Maven build configurations, handle token expiration states, and ensure smooth cross-role integration.

Testimonials from team

Team Member 1 :
"The project helped in improving analytical thinking and practical knowledge in modern frontend development, generative AI integration, and multi-role dashboard visualization while working on a real-world enterprise learning platform."
Team Member 2:
"Working on this project strengthened problem-solving abilities through tasks such as designing robust RESTful APIs in Spring Boot, managing relational database schemas in MySQL, and implementing secure JWT-based authentication workflows."
Team Member 3:
"The development of the SkillSphere platform enhanced technical skills in Java, React 19, database management, and AI-based system design while providing valuable experience in building unified, full-stack digital ecosystems."
Team Member 4:
"Designing the conversational AI Study Buddy and integrating external resources like the YouTube Data API helped me master client-side state management, responsive UI layout, and building reliable API failover models."
Team Member 5:
"This project provided hands-on experience in configuring secure role-based access control (RBAC), resolving CORS communication challenges between React and Spring Boot, and creating a seamless, unified user experience across the student, recruiter, and admin portals."

1. Conclusion

The **Enterprise Learning Platform** project successfully demonstrates the integration of full-stack web development and generative AI to create an intelligent learning and workforce management system. The platform allows students to follow personalized learning paths, practice in an interactive coding arena, build AI-assisted resumes, and search for job opportunities, while enabling recruiters to manage candidates and track hiring pipelines efficiently. The system also includes interactive, role-based dashboards, progress analytics, and a conversational AI Study Buddy to support users with instant learning guidance. Through this project, practical knowledge of Java, Spring Boot, React 19, MySQL database management, and Google Gemini API integration was applied to build a real-world enterprise application. Overall, the project highlights how modern technology and artificial intelligence can be leveraged to bridge the gap between education and industry, empowering students to develop career-ready skills and connect with employment opportunities.

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2. Acknowledgements

We express our sincere gratitude to **Infosys Springboard** for providing an excellent learning environment, industry-oriented curriculum support, and access to valuable resources that helped in the successful development of the **Enterprise Learning Platform** project.

We extend our heartfelt thanks to **Ms. Shakthi Gopalakrishnan** our mentor, for continuous guidance, constructive feedback, and encouragement throughout the project development process. The insights and suggestions provided played a significant role in improving our understanding and refining the implementation of the system.